

Micro Energy Harvesting 2025

Exercise 3: Vibrational Energy Harvesting

Task 1: Quality Factor

The quality factor Q of the harmonic oscillator is also defined as the following ratio:

$$Q = 2\pi \frac{\text{energy stored}}{\text{av. energy lost per period}} = 2\pi \frac{W}{P \cdot T} = \frac{W}{P/\omega_0}$$

Derive the quality factor from the kinetic and potential energy:

$$W = W_{Kin} + W_{Pot}, \quad W_{Kin} = \frac{1}{2} m \dot{x}^2, \quad W_{Pot} = \frac{1}{2} m \omega_0^2 x^2$$

using the equation of motion for the damped harmonic oscillation:

$$x = x_0 \cdot \underbrace{e^{-\frac{t}{2\tau}}}_{const.} \cdot \cos(\omega_0 \cdot t).$$

Hint: Assume that the exponential term is virtually constant for the time span of one periode.

Task 2: Radio protocol

Sending a radio protocol in a tunnel requires an assumed energy of $E_{\text{send}} = 200 \mu\text{J}$ (EnOcean STM300). Calculate the energy contained in the mass of $m = 1 \text{ g}$ with a resonant frequency of $f = 70 \text{ Hz}$ and an oscillation amplitude of $A_0 = 3 \text{ mm}$. How long would the energy last at a transmission frequency of $0,5 \text{ Hz}$? Assume that you can use 100% of the stored energy.

Task 3: Harvester Design

You want to design a harvester. First you measure the ambient vibration you find the dominant vibration modes: 100 Hz with $50 \mu\text{m}$ vibration amplitude, 200 Hz with $20 \mu\text{m}$ vibration amplitude and 2.5 kHz with $5 \mu\text{m}$ vibration amplitude. You have the choice between two harvesters with the stiffnesses $K_1 = 1 \text{ mN}/\mu\text{m}$ and $K_2 = 4 \text{ mN}/\mu\text{m}$. The parasitic damping constants are $c_{\text{dis1}} = 10 \text{ g/s}$ and $c_{\text{dis2}} = 200 \text{ g/s}$. The generator damping of harvester 1 can be chosen arbitrarily, while c_{harv2} can be chosen between $0.05 \text{ VAs}^2/\text{m}^2$ and $0.15 \text{ VAs}^2/\text{m}^2$.

- Choose the most suited set-up (harvester type 1 or 2, mass, charv). Justify your choice!
- What is the average stored power, if you assume a rectifier with 50% losses.
- Temperature changes lead to changes in material parameters and consequently resonance frequency. Which type of harvester is better in an environment with slightly varying temperature?